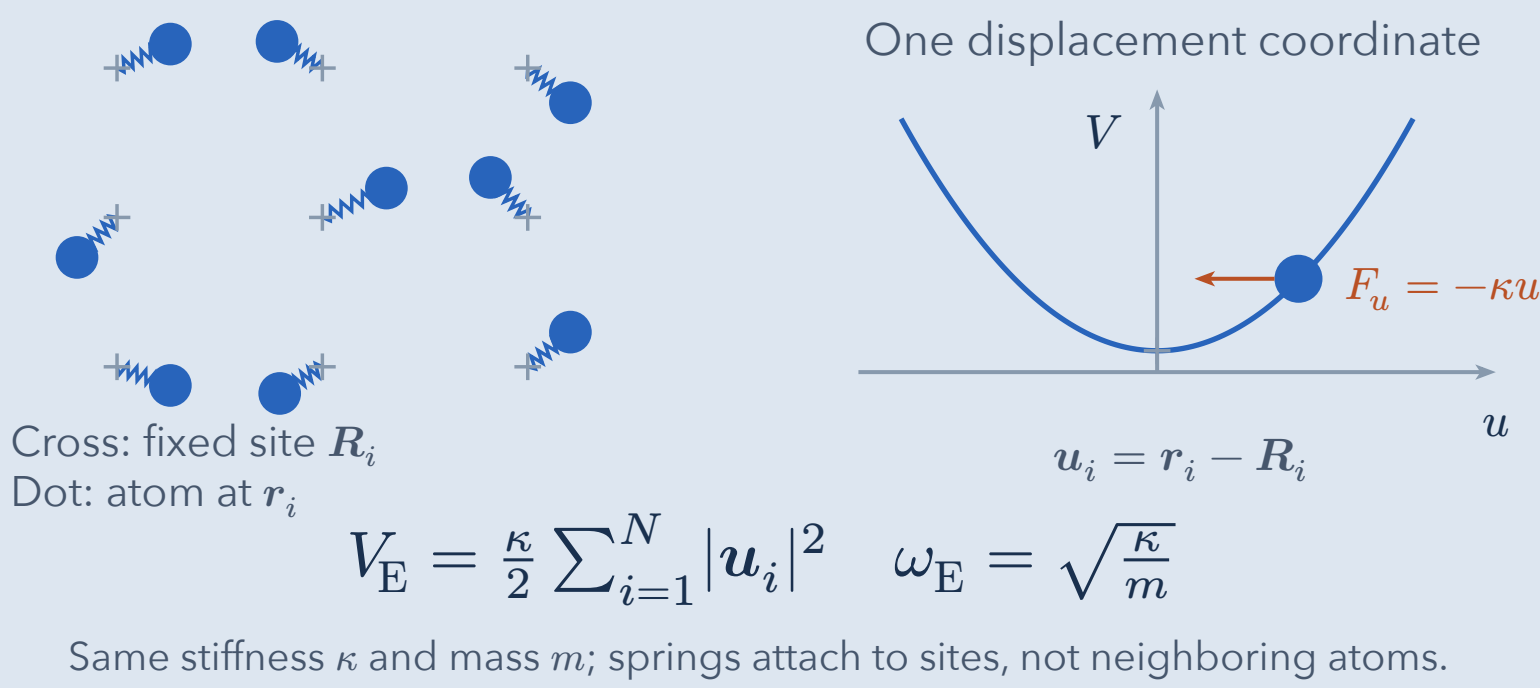


Einstein crystal

One solvable model connects atomic vibrations, quantum heat capacity, entropy, and solid free energies.

1 Replace the crystal by local springs

N atoms × 3 directions = M = 3N independent modes

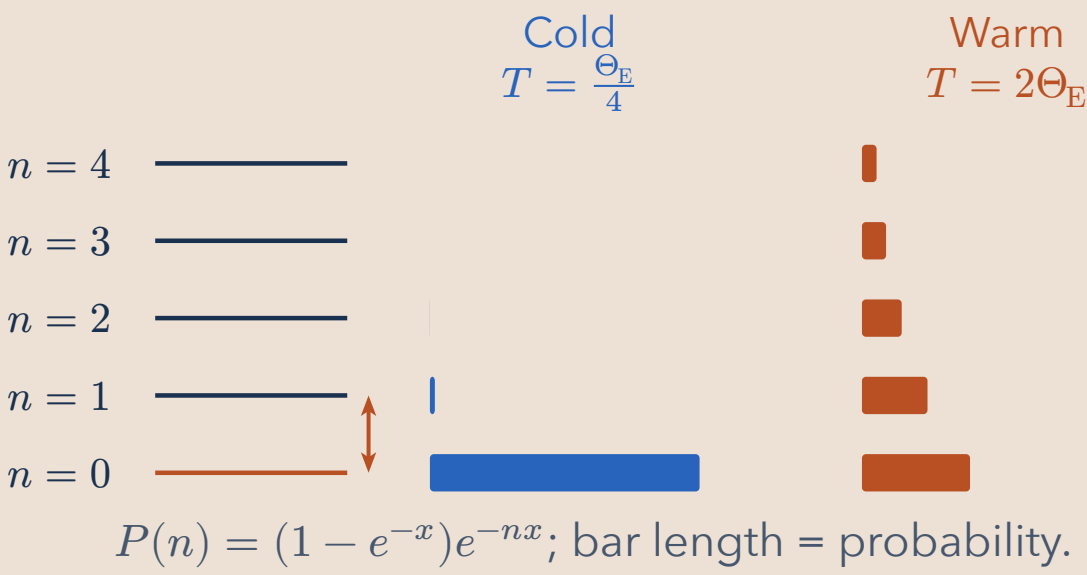


Read the restoring force

u_i is atom i 's displacement from its fixed site. The force points back toward the potential minimum: $F_u = -\kappa u$. Doubling $|u|$ quadruples the spring energy. Stiffer springs or lighter atoms have a higher angular frequency ω_E . This harmonic approximation describes small vibrations about stable sites.

2 Heating fills a quantum ladder

$$E_n = \left(n + \frac{1}{2}\right) \hbar \omega_E \quad n = 0, 1, 2, \dots$$



Gap $\hbar \omega_E$; the ground state still has $E_0 = \hbar \frac{\omega_E}{2}$. Higher levels omitted.

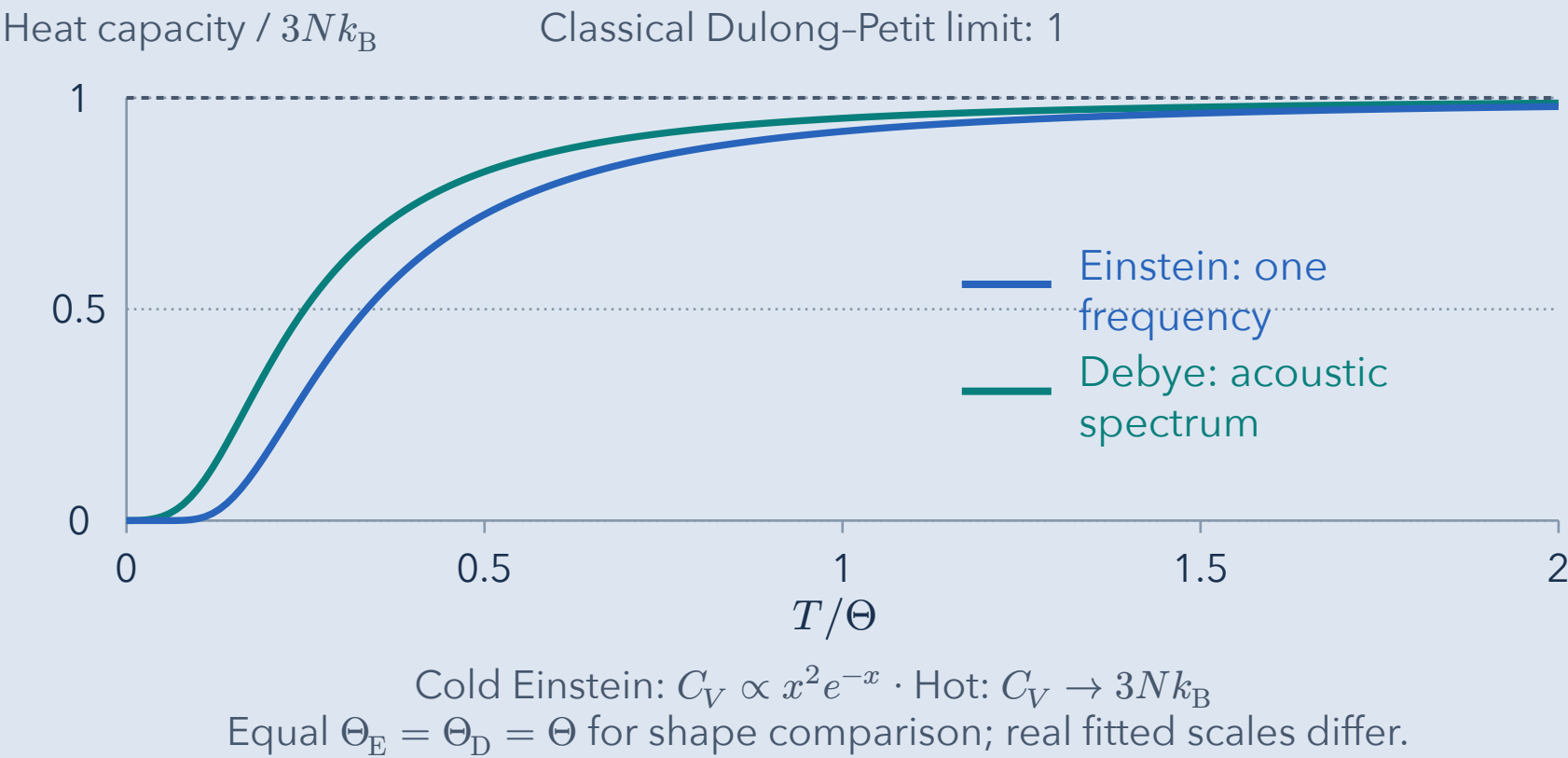
Thermal energy competes with the gap

n counts excitation quanta of one mode; $P(n)$ is its level probability. Θ_E expresses the energy gap as a temperature. Thus $x = \frac{\Theta_E}{T}$ compares the gap $\hbar \omega_E$ with thermal energy $k_B T$: large x means few excited states.

3 Why the heat capacity falls on cooling

Thermal excitations store heat. The fixed zero-point energy contributes no heat capacity.

$$\frac{C_V}{3Nk_B} = \frac{x^2 e^x}{(e^x - 1)^2} \quad x = \frac{\Theta_E}{T} \quad \Theta_E = \frac{\hbar \omega_E}{k_B}$$

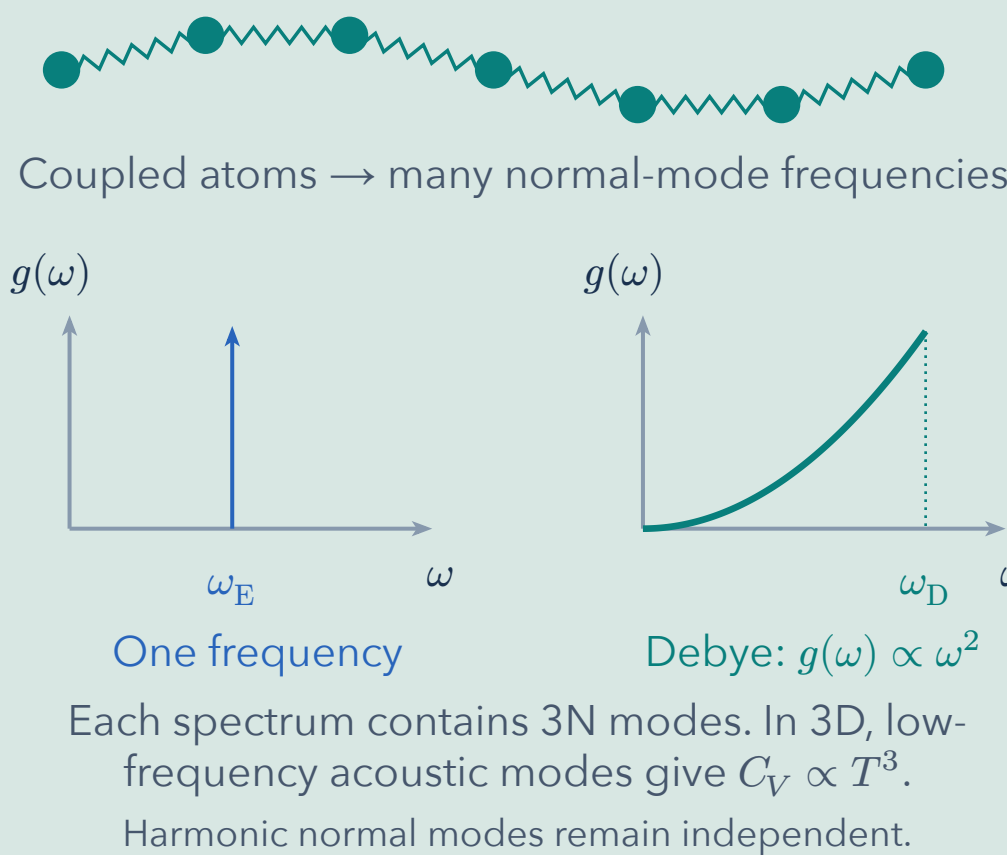


Heat capacity measures energy stored per degree

$C_V = \left(\frac{\partial U}{\partial T}\right)_V$ measures how much energy a small temperature rise stores at fixed volume. Classically, each mode contributes $\frac{k_B}{2}$ from motion and $\frac{k_B}{2}$ from its spring, so $3N$ modes give $3Nk_B$. At low T , a mode stores little extra heat when its excitation gap greatly exceeds $k_B T$.

4 Real crystals have phonons

Atoms move together. A phonon is one quantum of a collective vibration.

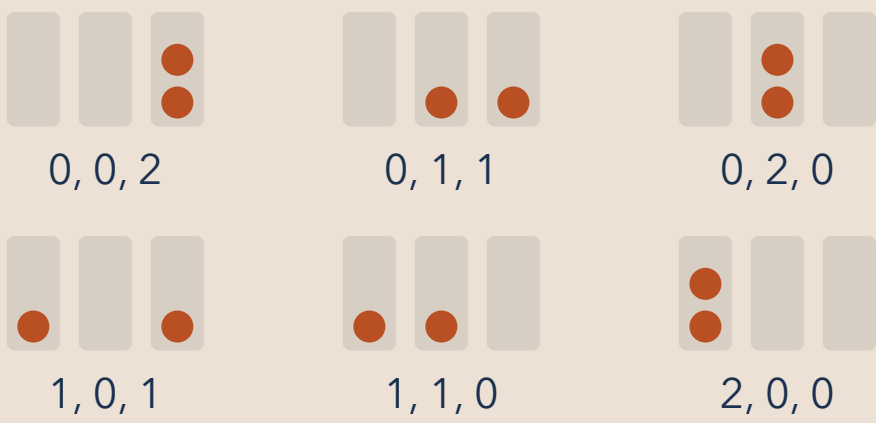


Why soft modes matter

$g(\omega) d\omega$ counts modes in a frequency interval. Soft acoustic modes remain easy to excite as the solid cools, giving $C_V \propto T^3$ at low T . ω_D sets the cutoff, with $\Theta_D = \hbar \frac{\omega_D}{k_B}$.

5 Count states to get entropy

One atom: $M = 3$ modes, $q = 2$ quanta. Six allocations have the same energy.



$$\Omega(M, q) = \binom{q+M-1}{q}$$

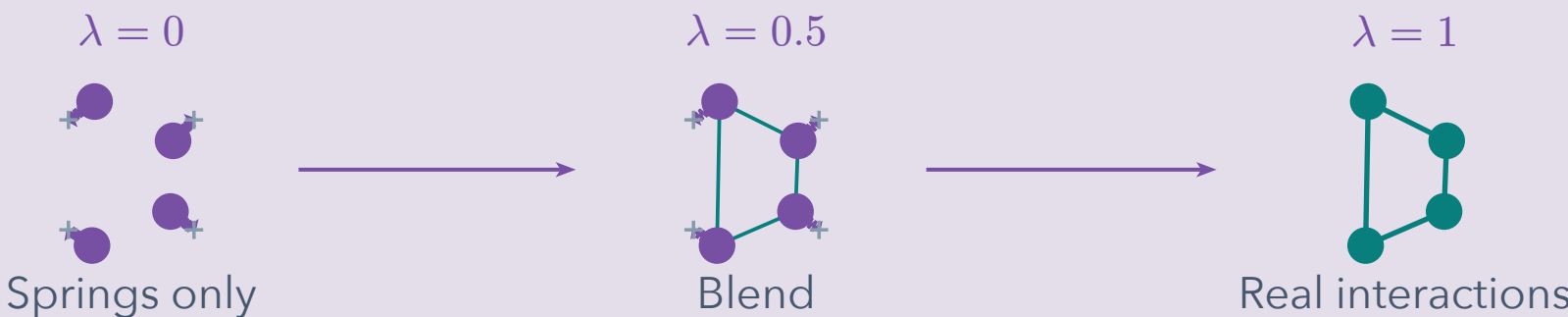
$$S = k_B \ln \Omega \quad M = 3N$$

Same energy, different microstates

Each bin is an x , y , or z mode. The total excitation energy is $q\hbar\omega_E$ above the ground state. Ω counts allocations at fixed q : more possibilities mean larger entropy. Here $\Omega = 6$ and $S = k_B \ln 6$.

6 A reference for solid free energies

Classical Frenkel-Ladd integration · Helmholtz free energy $F = U - TS$



$$V_\lambda = (1 - \lambda)V_E + \lambda V_{\text{real}}$$

$$F_{\text{real}} - F_E = \int_0^1 \langle V_{\text{real}} - V_E \rangle_\lambda d\lambda$$

$\langle \dots \rangle_\lambda$: equilibrium average in the blend. Keep temperature, volume, and the positional constraint fixed along a reversible crystal path. Release the fixed center of mass (or one particle: Einstein molecule) with its analytic free-energy correction.

Use: compare candidate crystal structures

$\lambda \in [0, 1]$ controls the blend. Sampling each value gives the average in the integral, which adds up the reversible free-energy change. The result includes both energy and entropy through $F = U - TS$. At the same temperature, volume, and particle number, the candidate with lower Helmholtz free energy F is favored.

Symbols: T temperature · k_B Boltzmann constant · $\hbar$ reduced Planck constant · C_V heat capacity at fixed volume · $g(\omega)$ mode density.

U internal energy · V potential energy · q excitation quanta. Heat curves include lattice vibrations only; no electronic or magnetic contribution. Thermal equilibrium is assumed.